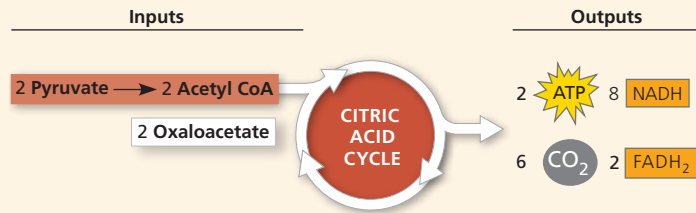


CONCEPT 9.3

After pyruvate is oxidized, the citric acid cycle completes the energy-yielding oxidation of organic molecules (pp. 171–174)

- In eukaryotic cells, pyruvate enters the mitochondrion and is oxidized to **acetyl CoA**, which is further oxidized in the citric acid cycle.

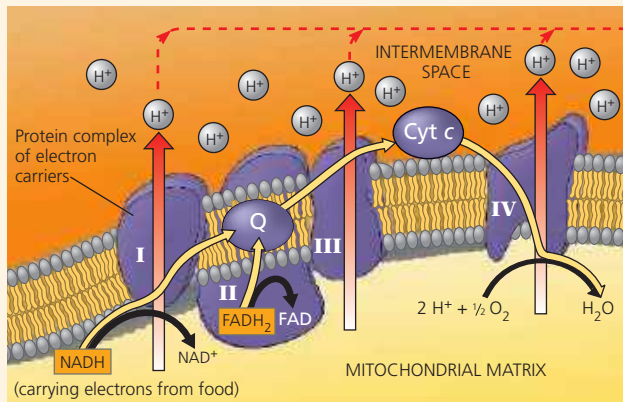


? What molecular products indicate the complete oxidation of glucose during cellular respiration?

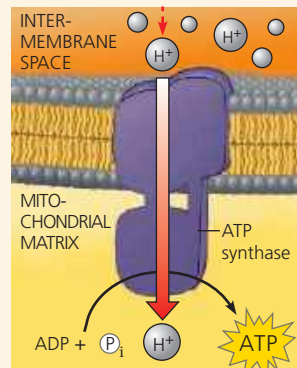
CONCEPT 9.4

During oxidative phosphorylation, chemiosmosis couples electron transport to ATP synthesis (pp. 174–179)

- NADH and FADH₂ transfer electrons to the electron transport chain. Electrons move down the chain, losing energy in several energy-releasing steps. Finally, electrons are passed to O₂, reducing it to H₂O.



- Along the electron transport chain, electron transfer causes protein complexes to move H⁺ from the mitochondrial matrix (in eukaryotes) to the intermembrane space, storing energy as a **proton-motive force** (H⁺ gradient). As H⁺ diffuses back into the matrix through **ATP synthase**, its passage drives the phosphorylation of ADP to form ATP, called **chemiosmosis**.
- About 34% of the energy stored in a glucose molecule is transferred to ATP during cellular respiration, producing a maximum of about 32 ATP.



? Briefly explain the mechanism by which ATP synthase produces ATP. List three locations in which ATP synthases are found.

CONCEPT 9.5

Fermentation and anaerobic respiration enable cells to produce ATP without the use of oxygen (pp. 179–182)

- Glycolysis nets 2 ATP by substrate-level phosphorylation, whether O₂ is present or not. Under anaerobic conditions, anaerobic respiration or fermentation can take place. In anaerobic respiration, an electron transport chain is present with a final electron acceptor other than oxygen. In fermentation, the electrons from NADH are passed to pyruvate or a derivative of pyruvate, regenerating the NAD⁺ required to oxidize more glucose. Two common types of fermentation are **alcohol fermentation** and **lactic acid fermentation**.
- Fermentation, anaerobic respiration, and aerobic respiration all use glycolysis to oxidize glucose, but they differ in their final electron acceptor and whether an electron transport chain is used (respiration) or not (fermentation). Respiration yields more ATP; aerobic respiration, with O₂ as the final electron acceptor, yields about 16 times as much ATP as does fermentation.
- Glycolysis occurs in nearly all organisms and is thought to have evolved in ancient prokaryotes before there was O₂ in the atmosphere.

? Which process yields more ATP, fermentation or anaerobic respiration? Explain.

CONCEPT 9.6

Glycolysis and the citric acid cycle connect to many other metabolic pathways (pp. 182–184)

- Catabolic pathways funnel electrons from many kinds of organic molecules into cellular respiration. Many carbohydrates can enter glycolysis, most often after conversion to glucose. Amino acids of proteins must be deaminated before being oxidized. The fatty acids of fats undergo **beta oxidation** to two-carbon fragments and then enter the citric acid cycle as acetyl CoA. Anabolic pathways can use small molecules from food directly or build other substances using intermediates of glycolysis or the citric acid cycle.
- Cellular respiration is controlled by allosteric enzymes at key points in glycolysis and the citric acid cycle.

? Describe how the catabolic pathways of glycolysis and the citric acid cycle intersect with anabolic pathways in the metabolism of a cell.

TEST YOUR UNDERSTANDING

➔ For more multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

Levels 1-2: Remembering/Understanding

- The *immediate* energy source that drives ATP synthesis by ATP synthase during oxidative phosphorylation is the
 - oxidation of glucose and other organic compounds.
 - flow of electrons down the electron transport chain.
 - H⁺ concentration gradient across the membrane holding ATP synthase.
 - transfer of phosphate to ADP.
- Which metabolic pathway is common to both fermentation and cellular respiration of a glucose molecule?
 - the citric acid cycle
 - the electron transport chain
 - glycolysis
 - reduction of pyruvate to lactate